

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Program (Matlab Code):

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---
figure;
```

```
freqz(b_rect,1);  
title('High-pass FIR using Rectangular Window');  
figure;  
freqz(b_hamm,1);  
title('High-pass FIR using Hamming Window');  
figure;  
freqz(b_hann,1);  
title('High-pass FIR using Hanning Window');
```

Command Window:

```
--- High-Pass Coefficients (Rectangular Window) ---  
-0.0089  
0  
0.0097  
0.0143  
0.0106  
0  
-0.0117  
-0.0175  
-0.0131  
  
0  
0.0148  
0.0225  
0.0171  
0  
-0.0202  
-0.0314  
-0.0247  
0  
0.0318  
0.0524  
0.0445  
0  
-0.0741
```

-0.1572
-0.2223
0.7407
-0.2223
-0.1572
-0.0741
0
0.0445
0.0524
0.0318
0

-0.0247
-0.0314
-0.0202
0
0.0171
0.0225
0.0148
0
-0.0131
-0.0175
-0.0117
0
0.0106
0.0143
0.0097
0
-0.0089

--- High-Pass Coefficients (Hamming Window) ---

-0.0007
0
0.0009
0.0016
0.0015
0
-0.0024
-0.0044
-0.0039
0

0.0060
0.0103
0.0088

0
-0.0128
-0.0217
-0.0184
0
0.0268
0.0464
0.0410
0
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0

-0.0007

--- High-Pass Coefficients (Hanning Window) ---

-0.0000

0

0.0003

0.0008

0.0009

0

-0.0020

-0.0038

-0.0035

0

0.0057

0.0100

0.0087

0

-0.0127

-0.0216

-0.0183

0

0.0267

0.0464

0.0410

0

-0.0726

-0.1568

-0.2243

0.7500

-0.2243

-0.1568

-0.0726

0

0.0410

0.0464

0.0267

0

-0.0183

-0.0216

-0.0127

0

0.0087

0.0100

0.0057

0

-0.0035

-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000

PLOTS:



